TT-R-251J
August 1, 1978
SUPERSEDING
Fed. Spec. TT-R-251H
April 26, 1967

FEDERAL SPECIFICATION

REMOVER; PAINT (ORGANIC SOLVENT TYPE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers four types of paint removers commonly employed in scrape-off operations, and suitable for use on finished wood and metal surfaces (see 6.1).

1.2 Classification.

1.2.1 Types and classes. Paint removers covered by this specification shall be of the following types and classes, as specified (see 6.1).

Type I - Flammable mixture with paintable retardant.

Type II - Flammable mixture with nonpaintable retardant.

Type III - Nonflammable mixture with paintable retardant.
Class A - Low viscosity.
Class B - High viscosity.

Type IV - Nonflammable mixture, water rinsable with paintable retardant.
Class A - Low viscosity.
Class B - High viscosity.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-E-489 - Enamel, Alkyd, Gloss (For Exterior and Interior Surfaces.)
TT-P-636 - Primer Coating, Alkyd, Wood and Ferrous Metal.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-C-636 - Boxes, Shipping, Fiberboard.

Federal Standard:

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 1364 - Water in Volatile Solvents (Fischer Reagent Titration Method.)
D 1545 - Viscosity of Transparent Liquids by Bubble Time Method.
D 3278 - Flash Point of Liquids by Setaflash Closed Tester.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American National Standards Institute (ANSI) Standards:

MH2.1 - 55-Gallon Tight-Head Universal Drums (DOT-17E).
MH2.3 - 55-Gallon Tight-Head Universal Drum (DOT-5B).
MH2.5 - 55-Gallon Full-Removable-Head Universal Drum (DOT-17H).
MH2.15 - 5-Gallon Nesting Lug-Cover Universal Pail (DOT-37A80, DOT-37A60, UFC-Rule 40, NMFC - Item 260).

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Volatile solvent. The solvent, when tested as specified in 4.4.14, shall conform to the following requirements by volume:

(a) The total of solvents with olefinic or cyclo-olefinic unsaturation shall not exceed 5 percent.
(b) The total of aromatic compounds with 8 or more carbon atoms in the molecule, except ethylbenzene, methyl benzoate, and phenyl acetate, shall not exceed 8 percent.
(c) The total of ethylbenzene, toluene, and branched-chain ketones shall not exceed 20 percent.
(d) A solvent which may be classified into more than one of the above groups shall be considered a member of the group having the lowest allowable concentration.
(e) The total of (a), (b), and (c) shall not exceed 20 percent.

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3.2 Quantitative requirements. The quantitative requirements shall be as specified in table I.

TABLE I. Quantitative requirements*

Characteristics	Type I		Type II		Type III				Type IV			
					Class A		Class B		Class A		Class B	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Nonvolatile matter (percent)	--	7.0	--	15.0	--	7.0	--	10.0	--	10.0	--	10.0
Ash (percent)	--	0.3	--	0.3	--	0.3	--	0.3	--	0.7	--	0.7
Fixed alkalies	--	**	--	**	--	**	--	**	--	**	--	**
Viscosity (Gardner bubble viscometer)		B tube		B tube		B tube		C tube		B tube		C tube
Loss by evaporation (percent)	--	28.0	--	5.0	--	12.0	--	5.0	--	12.0	--	5.0
Water content (percent)	--	--	--	--	--	0.5	--	0.5	--	0.5	--	0.5
Methylene chloride (percent)	--	--	--	--	70	--	70	--	70	--	70	--
Carbon tetrachloride (percent)	--	0.3	--	0.3	--	0.5	--	0.5	--	0.5	--	0.5
Phenol (percent)	--	0.2	--	0.2	--	0.2	--	0.2	--	0.2	--	0.2
Total cresols (percent)	--	0.1	--	0.1	--	0.1	--	0.1	--	0.1	--	0.1

* In this table, percent refers to percent by weight of paint remover.
** No pink color.

3.3 Qualitative requirements.

3.3.1 Condition in container. The remover shall not settle, cake, gel, or thicken to such a degree it cannot be redispersed easily to a uniform and homogeneous mixture, when tested as specified in table II.

3.3.2 Removing performance. The material, when applied to painted surfaces and tested as specified in 4.4.8, shall completely remove the coating.

3.3.3 Paintable property (types I, III, and IV). Following the stripping operation and recoating of the panel, the resultant coating shall not flake or chip when tested as specified in 4.4.9.

3.3.4 Temperature stability. The material, when tested in accordance with 4.4.10, shall be as readily dispersed to its original appearance and consistency as the original portion (neither heated nor chilled).

3.3.5 Effect on wood. The material, when tested as specified in 4.4.11, shall not stain the test panels or cause the grain to raise.

3.3.6 Effect on metal. The material shall not rust or corrode when tested as specified in 4.4.12.

3.3.7 Rinsability (type IV only). When tested as specified in 4.4.13, the coating that has been lifted by the remover shall be completely removed by flushing with a stream of water.

3.3.8 Flash point. Types 3 and 4 paint remover shall have no flash point when tested as specified in table II.

3.3.9 Precautionary marking.

3.3.9.1 The precautionary marking shall contain the following (see 5.2).

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"Flammable, volatile solvent." (Types I and II only)
"Use with adequate ventilation."
"Avoid prolonged breathing of vapors or spray mists."
"Protective equipment, such as gloves and face shield, shall be worn when working with this paint remover."

3.4 Quantities. The paint remover shall be furnished in 1-quart, 1-gallon, 5-gallon, and 55-gallon quantities.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of section 5. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

4.3 Sampling of the end item for testing. For the purposes of sampling, a lot shall be the end product of all raw materials mixed, blended, or processed in a single operation, and shall be expressed in units of gallons. Samples from lots shall be taken in accordance with MIL-STD-105 using inspection level S-2 and an acceptable quality level (AQL) of 1.5.

4.4 Test procedures. The paint remover shall be tested according to the test methods listed in table II. Unless otherwise specified, standard testing conditions are a temperature of 23°C ± 1°C (73°F ± 2°F) and a relative humidity of 50 ± 5 percent. All test reports shall contain the individual values utilized in expressing the final result. All tests shall be evaluated for conformance to the requirements specified in section 3. Failure to pass any test, or noncompliance with any requirement, shall be cause for rejection of the sample.

TABLE II. Test methods

Characteristic	Required reference	Applicable Test Method: Fed. Test Method Std. No. 141	ASTM Method	Paragraph reference
Nonvolatile matter	Table I	----	----	4.4.1
Ash	Table I	----	----	4.4.2
Fixed alkalies	Table I	----	----	4.4.3
Viscosity	Table I	----	D 1545	-----
Loss by evaporation	Table I	----	----	4.4.4
Water content	Table I	----	D 1364	-----
Methylene chloride	Table I	----	----	4.4.5
Carbon tetrachloride	Table I	----	----	4.4.6
Phenol	Table I	----	----	4.4.7
Cresols	Table I	----	----	4.4.7
Condition in container	3.3.1	3011	----	----
Removing performance	3.3.2	----	----	4.4.8
Paintable property (types I, III, IV)	3.3.3	----	----	4.4.9
Temperature stability	3.3.4	----	----	4.4.10
Effect on wood	3.3.5	----	----	4.4.11
Effect on metal	3.3.6	----	----	4.4.12
Rinsability (type IV)	3.3.7	----	----	4.4.13
Flash point	3.3.8	----	D 3278	----
Solvent analysis	3.1	7356	----	4.4.14

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4.4.1 Nonvolatile matter. Place a portion of the thoroughly-mixed remover in a stoppered bottle or weighing pipette. Weigh container and sample. Transfer 5.0 ± 0.1g of the sample to a weighed crucible. Weigh container again and by difference calculate the exact weight of the sample used. Heat the crucible on a steam bath for 30 minutes; then in an oven at 105°C ± 2°C for one hour, cool and weigh. From the weight of the residue left in the crucible and weight of the sample, calculate the percentage of nonvolatile matter for compliance with the requirement in table I.

4.4.2 Ash. Heat the crucible containing the nonvolatile matter (obtained in 4.4.1) until it is burned off, using only moderate heat not to exceed 550°C; cool and weigh. From the weight of the residue left in the crucible and weight of the sample (see 4.4.1), calculate the percentage of ash for compliance with the requirement in table I. Save the ash for the fixed alkali test.

4.4.3 Fixed alkalies. Dissolve the soluble portion of the ash (4.4.2) in 25 ml of distilled water. Add 3 or 4 drops of phenolphthalein indicator solution, and note any change in color for compliance with the requirement in table I.

4.4.4 Loss by evaporation. Place a portion of the thoroughly-mixed remover in a stoppered bottle or a weighing pipette. Weigh container and sample. Transfer 5.0 ± 0.1 g of the sample to a weighed flat-bottomed dish (from 60 to 70 mm in diameter). Allow the dish and contents to stand undisturbed and protected from air currents or drafts at standard conditions for 30 minutes. Reweigh the dish and contents, and from the loss in weight and the weight of the sample, compute the percentage loss by evaporation for compliance with the requirement in table I.

4.4.5 Methylene chloride (types 3 and 4).

4.4.5.1 Standard. Prepare 50 ml of a standard mixture of methylene chloride containing small amounts of carbon tetrachloride, phenol, and cresols such that the concentrations of each ingredient are accurately known to the nearest 0.1 percent (e.g. - 95.8 percent methylene chloride, 0.8 percent phenol, 0.4 percent p-cresol, and 1.0 percent each of carbon tetrachloride, m-cresol, and o-cresol).

4.4.5.2 Equipment. Use a Hewlett-Packard Model 5754 Gas Chromatograph (or equal) equipped with an integrator. Use a 1.83 m (6 ft), 3.18 mm (1/8 in) o.d. stainless steel column packed with 10 percent of high molecular weight dimethyl silicone suitable for chromatography (such as SE-30, OV-1 or equal) on 80-100 mesh Chromosorb W (AW-DMCS) or equal.

4.4.5.3 Procedure. Use a temperature program for the gas chromatograph between 100-150°C at 6°C/min with a 2 minute post injection hold at 100°C and a 5 minute upper limit hold at 150°C. Set the flame ionization detector at 175°C and the injection port at 175°C. Set the attenuation so all peaks remain on scale. Test the standard mixture prepared in 4.4.5.1 and obtain the apparent concentration (as recorded by the integrator) for each corponent. Then test the sample, and obtain the apparent concentration (as recorded by the integrator) for each component.

4.4.5.4 Calculation.

$$\text{true conc. } CH_2Cl_2 \text{ in sample} = \frac{\text{true conc. } CH_2Cl_2 \text{ in std. mix}}{\text{apparent conc. } CH_2Cl_2 \text{ in std. mix}} \times \text{apparent conc. } CH_2Cl_2 \text{ in sample}$$

The true concentration of CH_2Cl_2 in the sample shall be evaluated for compliance with the requirement in table I.

4.4.6 Carbon tetrachloride.

4.4.6.1 Method of determination. Use the same standard mixture, equipment, and procedure as specified in 4.4.5.1, 4.4.5.2, and 4.4.5.3, respectively.

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4.4.6.2 Calculation.

$$\text{true conc. } CCl_4 \text{ in sample} = \frac{\text{true conc. } CCl_4 \text{ in std. mix}}{\text{apparent conc. } CCl_4 \text{ in std. mix}} \times \text{apparent conc. } CCl_4 \text{ in sample}$$

The true concentration of CCl_4 in the sample shall be evaluated for compliance with the requirement in table I.

4.4.7 Phenol and cresols.

4.4.7.1 Method of determination. Use the same standard mixture, equipment and procedure as specified in 4.4.5.1, 4.4.5.2, and 4.4.5.3, respectively.

4.4.7.2 Calculation.

$$\text{true conc. M in sample} = \frac{\text{true conc. M in std. mix}}{\text{apparent conc. M in std. mix}} \times \text{apparent conc. M in sample}$$

Where M = phenol or o-, p-, or m-cresol.

Evaluate for compliance with the requirement in table I using the true concentration of M in the sample and verify according to 4.4.8.3 that phenol and/or cresols are present.

4.4.7.3 Verification method. If peaks for phenol and/or cresols are identified by the method in 4.4.7.1, an additional qualitative test shall be run to verify that these peaks are actually phenol and cresols. The samples for this procedure shall be the paint remover by itself and a mixture of 1 part of the standard (as specified in 4.4.5.1) in 9 parts of the paint remover. The equipment shall be as specified in 4.4.5.2. The gas chromatograph temperature shall be isothermal at 70°C, with the flame ionization detector and the injection port at 175°C. The column temperature should be raised to 150°C between runs for 5-10 minutes to permit higher boiling components to come off and not interfere with the next run. The chromatograms should develop in about 30 minutes at 70°C. The sensitivity of the instrument shall be set high enough so the smaller peaks are readily distinguished from the baseline. Compare chromatograms for the mixture and the paint remover alone to verify the presence of phenol and cresols. Peaks shall be matched between the two chromatograms, and any extra peaks appearing in the chromatogram for the remover plus standard belong to components that are present in the standard but not in the remover.

4.4.8 Removing performance.

4.4.8.1 Test panel preparation. Steel test panels shall conform to method 2011 of Fed. Test Method Std. No. 141 and shall be thoroughly cleaned by solvent spray according to procedure A of method 2011.

4.4.8.2 Test coatings preparation.

4.4.8.2.1 For types I and II removers, coat the panel by brushing or spraying with white enamel conforming to TT-E-489, class A. Each coat shall have a dry film thickness of 0.025 ± 0.0005 mm (0.001 ± 0.0002 in), and the dry film thickness of the system shall be 0.076 ± 0.005 mm (0.003 ± 0.0002 in). Allow 24 hours dry time between coats.

4.4.8.2.2 For types III and IV removers, prepare a coating system by either brushing or spraying as follows: Apply a coat of primer conforming to TT-P-636 to the panel, allow to air dry in a horizontal position at standard condition for 4 hours, and then bake at 105°C ± 2°C for 18 hours. Over the dried primer, apply a coat of white enamel conforming to TT-E-489, class A; allow to air dry for 4 hours; and bake for 18 hours. Apply a second coat of the white enamel; allow to air dry for 4 hours; and bake the system for 18 hours at 105°C ± 2°C. Each coat shall have a dry film thickness of 0.025 ± 0.005 mm (0.001 ± 0.0002 in).

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4.4.8.3 Removal of test coating. Thoroughly mix the remover, which shall have been kept at standard conditions for at least 4 hours. Apply a layer of 51 mm (2 in) width and 0.25 mm (.010 in) depth of the remover to the test coating, using a doctor blade or other suitable device. Allow the test specimen to remain in a horizontal position, protected from drafts for 45 minutes. Remove the loosened coating from the metal panel with rubber plate scraper. The scraper should be held at an angle of 45° and moved across the face of the panel. Sponge off the loose pieces of coating and the residual remover with a pad of cheesecloth wetted with organic solvent. Dry with a clean, soft cloth, and note if the coating has been completely removed from the 51 mm wide strip to which the remover was applied in order to comply with the requirement in 3.3.2.

4.4.9 Paintable property (types I, III, and IV). Strip the coating from a panel used in the performance test (4.4.8), but do not sponge off the residual remover with organic solvent. Refinish the panel with the same coating. The adhesion of the new finish shall be determined in accordance wtih method 6304 of Fed. Test Method Std. 141 and compared with the finish of a similarly painted new panel prepared at the same time as the refinished panel to determine compliance with the requirement in 3.3.3.

4.4.10 Temperature stability. Thoroughly mix the remover and immediately fill three 150 mm test tubes to within 50 mm of the top with portions therefore and tightly stopper the tubes. Place one of the tubes in a water bath at a temperature of 38°C ± 2°C + and allow to remain for one hour. Place the second tube in a refrigerator maintained at -10°C ± 1°C for one hour. At the end of one hour bring all three tubes to standard conditions and evaluate for compliance with the requirement in 3.3.4.

4.4.11 Effect on wood. On smoothly sanded panels of maple, oak, and pine wood place 10 drops of the thoroughly-mixed remover. Allow the remover to remain in contact with the wood for 45 minutes. Remove that portion of remover remaining on the wood using first, filter paper to blot off the excess, and then lightly sponging the affected area with turpentine. Allow the panel to remain at standard conditions for one hour and examine for compliance with the requirement in 3.3.5.

4.4.12 Effect on steel. Use three steel strips prepared as specified in 4.4.8.1 and measuring 19 x 127 mm (3/4 x 5 in). Immerse one of the strips in the remover, held in the test tube, to approximately 4/5 the length of the strip. Stopper the tube and place it in an oven at 38°C ± 1°C for 24 hours. Immerse another strip to 4/5 its length in the remover and immediately remove it. Allow this strip to drain and dry in a vertical position for 6 hours at standard conditions. At the end of the respective exposure periods, clean the adhering remover and residue from the strips with solvent. Examine and compare the three strips, and evaluate for compliance with the requirement in 3.3.6.

4.4.13 Rinsability (type IV only). Two panels shall be prepared and the remover applied as specified in 4.4.8. Allow the test specimens to remain in a horizontal position protected from drafts for 45 minutes. Then rinse both panels with tapwater. The tapwater shall be at a temperature of 24°C ± 3°C and shall flow vertically downward through a 51 mm (2 in) length of glass tubing having an inside diameter of 6.4 mm (1/4 in). The rate of flow shall be 2 gallons of water per minute and the duration of rinsing shall be 1 minute. The rinsing shall be accomplished while moving the panel back and forth in a horizontal plane 51 mm (2 in) below the glass tubing outlet. Light brushing or rubbing shall be used to start emulsification. Evaluate for compliance with the requirement in 3.3.7.

4.4.14 Solvent analysis.

4.4.14.1 Solvent composition. The solvent composition shall be determined in accordance with 4.4.14.2 and method 7356 of Fed. Test Method Std. No. 141 to determine compliance with the requirement of 3.1.1.

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4.4.14.2 Aromatic and oxygenated solvents. The 1.83 m (6 ft) column shall be installed and the operating conditions described in method 7356 shall be followed. About 3 microliters of the isolated distillate shall be injected and the chromatogram scanned. The aliphatic solvents will emerge within 1 minute and the complete chromatogram should develop in about 5 minutes. From the position of the peaks observed on the chromatogram, an internal standard that will be free of interference shall be selected, such as cyclopentanol or cyclohexanol. Six-tenths of a ml of internal standard shall be added to 3 ml of the distillate. The sample shall be analyzed according to the above procedure. Peaks emerging after 1 minute are aromatic and oxygenated solvents shall be calculated as follows:

$$\text{percent aromatic and oxygenated solvent, v/v} = \frac{A \times B}{C \times D}$$

Where A = Percent of internal standard added (in this case, 20).
B = Area of aromatic and oxygenated solvents.
C = Calibration factor for the internal standard. This factor is dependent on the internal standard used and on the performance of the chromatograph, and should be determined daily.
D = Area of the internal standard (in this case cyclopentanol or cyclohexanol).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. The 1-quart and 1-gallon quantities of remover shall be packaged in metal cans conforming to PPP-C-96, type V, class 4. Metal screw cap closure is required. Inner friction metal seals are required and shall be rolled or tapped firmly into the screw neck, after cans have been filled, and seated so as not to damage liner pad or facing. Caps shall be secured by mechanical means or by cap wrenches. External plan B coating and side seam striping are required. One-gallon containers shall be provided with a formed bridge type handle or a wire handle, which shall be galvanized or otherwise protectively coated to resist corrosion. See paragraph 5.2.1 for the preparation for delivery of 5-gallon and 55-gallon quantities.

5.1.2 Commercial. The 1-quart and 1-gallon quantities of remover shall be packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling and storage. See paragraph 5.2.2 for the preparation for delivery of 5-gallon and 55-gallon quantities.

5.2 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.2.1 Level A. Twelve 1-quart cans, or six 1-gallon cans of remover, packaged as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-636, grades V3C, V3s, or V2s. The boxes shall be closed, waterproofed, and reinforced in accordance with the appendix of PPP-B-636. Alternatively, wirebound, cleated plywood, or nailed wood boxes shall be acceptable shipping containers when lined with a waterproof barrier material. The barrier material shall be sealed at the edges with waterproof tape or adhesive. The 5-gallon quantities of remover shall be furnished in metal pails conforming to ANSI Standard MH2.15, 24-gage. The 55-gallon quantities of remover shall be furnished in metal furnished in metal drums conforming to ANSI Standards MH2.1, Type I, MH2.3, or MH2.5.

5.2.2 Commercial. The remover shall be packed in a manner that will assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling and storage. The shipping container shall be in compliance with the National Motor Freight Classification or Uniform Freight Classification.

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5.3 Palletization. When shipments to Government depots are full car or truckload, the shipping containers shall be palletized on expendable wooden pallets, 2 or 4 way design, to facilitate handling in accordance with normal commercial practice. The palletized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.4 Marking. Marking shall be as specified in the contract or order.

6. NOTES

6.1 Intended use. Paint removers specified herein are generally intended for use indoors, but they may be used out of doors, except in hot or breezy conditions. Type I and II, flammable removers, are intended for use in places that are free from fire hazards. Types III and IV, nonflammable removers, are intended for use where a fire hazard may be encountered. Any type should be capable of removing air-dried, single coat films of oleoresinous and alkyd coatings, but may not be effective with newer synthetic-based coatings. The type III removers are considered more satisfactory for removing baked-on, air-dried, multiple-coat systems; however, they also may not be effective with some newer coatings, such as epoxies and polyurethanes. Types I and III removers may be used with solvent cleaning prior to refinishing, but this does not apply to type II. The class B removers are recommended where vertical surfaces are to be treated. Type II (nonpaintable) removers may contain wax which must be thoroughly removed before repainting. Type IV removers contain emulsifiers, and shall be used in those operations requiring water-rinsable properties. In such operations, flushing should be accompanied by vigorous scrubbing with a stiff brush.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) Level of packaging and packing required (see 5.1).
(d) Size and type of container required (see 5.1).
(e) Marking required (see 5.1 and 5.2).

6.3 Shelf-life surveillance. For the purpose of shelf-life surveillance, the paint remover stored in the original, unopened container shall meet the requirements in 3.3.1, 3.3.2, and the viscosity specified in table I.

Preparing Activity:

GSA - FSS

Military Coordinating Activity:

Army - MR

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 70 cents each.

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